

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2025)

Quiz 2

Total Marks: 30

Time: 45 Minutes

General Guidelines

- Attempt any three questions only.
- No cheat sheet is allowed.
- Show all steps clearly; unsupported answers may not receive full credit.

Question 1. [10 marks] Prove the following statements.

- (a) [7 marks] A necessary condition for a series $\sum_{n=1}^{\infty} u_n$ of real numbers to be convergent is that the corresponding sequence $(u_n)_{n=1}^{\infty}$ converges to 0.
- (b) [3 marks] The condition $\lim_{n \rightarrow \infty} u_n = 0$ is not sufficient for the series $\sum_{n=1}^{\infty} u_n$ of real numbers to be convergent.

Question (2). [10 Marks] Answer following questions.

- (a) [4 Marks] Prove or disprove that the series $\sum_{n=1}^{\infty} \frac{(n+1)^{1/n}}{n^2}$ is convergent.
- (b) [6 Marks] Prove or disprove that the series $\sum_{n=1}^{\infty} a_n$, $a_n = \frac{1}{2^n} \left(1 + \frac{(-1)^n}{n} \right)$ is convergent. If convergent, compute the sum of the series.

Question 3. [10 marks] Let $\sum_{n=1}^{\infty} u_n$ be a series of positive real numbers and suppose that

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l.$$

Then prove that

- (a) [5 marks] The series $\sum_{n=1}^{\infty} u_n$ is convergent if $l < 1$;
- (b) [5 marks] The series $\sum_{n=1}^{\infty} u_n$ is divergent if $l > 1$.

Question 4. [3+6+1 Marks] Consider the series

$$\sum_{n=3}^{\infty} \frac{1}{(n+2)(n+4(-1)^n)}.$$

Show that the series is convergent/divergent. If it is convergent, then compute the sum. Can this series be convergent but not absolutely convergent?